



Auralis

Forest Species Detection from Audio

DSP FINGERPRINTS / TRAINED CNN / PRETRAINED PANN

September 26, 2026

Problem statement



Input

A forest audio recording of arbitrary duration, format, sample rate, and recording quality. An optional site/location tag restricts the candidate species list; it is a categorical filter, not a continuous model feature.

Output

A confidence for every candidate species, thresholded into the final list of species judged present.

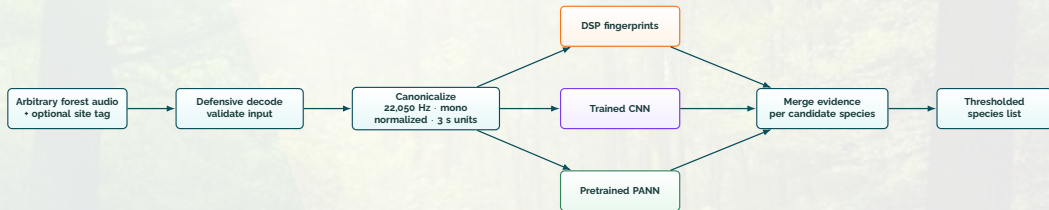
Signals-lab constraint

The team implements resampling, normalization, filtering, windowing, and spectrogram generation rather than outsourcing the DSP chain to black-box libraries.

Reliability requirement

The pipeline must safely accept unfamiliar formats, rates, durations, and quality levels. Species outside the trained label space are a model-coverage limitation, not an ingestion failure.

Proposed solution



Why three paths?

DSP finds recurring patterns, the project CNN learns target classes, and PANN contributes broad pretrained knowledge. Their different failure modes strengthen the ensemble.

Site-aware filtering

Location removes biologically impossible candidates before final ranking. It never substitutes for acoustic evidence or enters the model as a numeric feature.

Canonical audio contract



Every processing unit

Sample rate	22,050 Hz
Channels	Mono; average down-mix
Loudness	Fixed reference level
Duration	3 seconds
Native formats	WAV and FLAC
Optional format	MP3 via decoder

One boundary

Training and uploaded audio use the same canonicalization. Long files become overlapping 3-second units; short units are handled consistently.

Why it matters

Fixed rate, channel count, scale, and duration prevent a classifier from mistaking accidental input differences for biological evidence.

Documented exceptions

MP3 uses a standard call only for container/codec parsing; lossy compression may already have removed high-frequency evidence. PANN resamples its canonical copy to its model-required 32 kHz.

Robust audio handling

- ▶ WAV/FLAC accepted natively; MP3 decoding optional.
- ▶ Team-built resampling, downmixing, normalization, filtering, windowing, and spectrograms.
- ▶ Sliding 3-second units support arbitrary recording lengths.
- ▶ Unfamiliar rates and quality levels are canonicalized before inference.

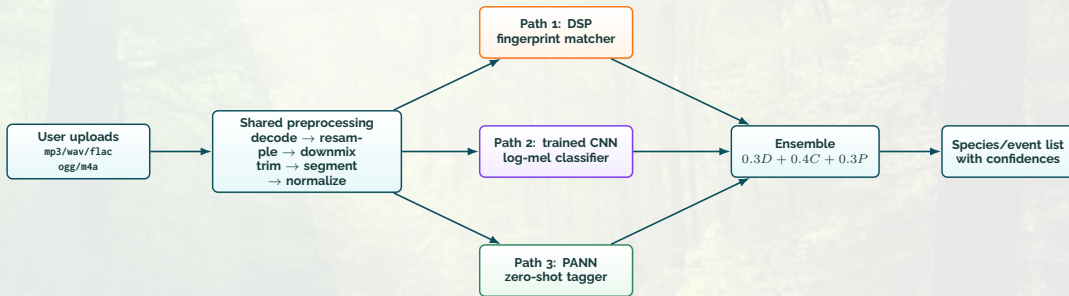
Detection behavior

- ▶ Confidence per candidate species, followed by thresholding.
- ▶ Optional location filtering narrows biological candidates.
- ▶ DSP, CNN, and PANN failures are isolated and reported rather than crashing the full analysis.
- ▶ Multi-path evidence supports species and relevant acoustic-event detection.

Scope boundary

Robust ingestion does not imply recognition of a species absent from every model's label space. Extending biological coverage is a data/model task.

The big picture



	DSP	CNN	PANN
Nature	Hand-built signal matching	Supervised model trained from scratch	Pretrained CNN14/AudioSet
Training Question	References only "Does this pattern reappear?"	50 epochs on synthetic data "Which class looks similar?"	No fine-tuning "What does AudioSet know?"

Shared preprocessing: one upload, consumer-specific outputs



AudioPreprocessor

Driven by a ConsumerSpec: target sample rate, channels, window duration/hop, and amplitude range. One raw upload is forked into the exact representation each path needs.

- ▶ Trim at 35 dB below peak; keep internal pauses.
- ▶ DSP receives the whole clip; neural paths receive overlapping windows.
- ▶ Peak normalization targets each consumer's expected range.

Shared convolution engine

FIR resampling and bandpass filtering use the convolution theorem:

$$x * h = \text{IFFT}(\text{FFT}(x) \text{FFT}(h)).$$

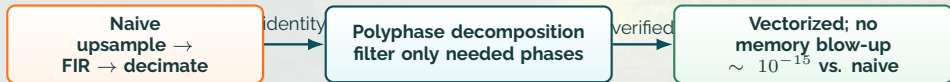
Decoding tries soundfile first and uses librosa.load as the fallback for compressed containers.

Original failure

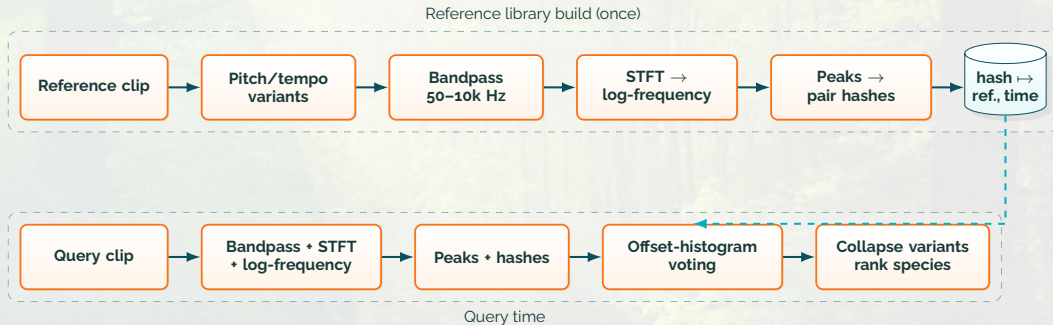
A windowed-sinc low-pass FIR supplies anti-aliasing. Naively upsampling by L , filtering, then decimating by M creates an intermediate array L times longer than the input. For $22,050 \rightarrow 48,000$ Hz, $L = 320$ —large enough to trigger an out-of-memory kill on a real clip.

Polyphase fix

1. Reduce L/M with Fraction; hence $\gcd(L, M) = 1$.
2. Split the FIR into L short phase filters.
3. Convolve phases directly with the original signal.
4. Fold decimation into phase selection using $M^{-1} \bmod L$.



Pathway 1: DSP fingerprint matcher



Inspired by acoustic fingerprinting: identify reproducible constellations of spectrogram peaks, then demand temporal consistency rather than mere spectral similarity.

DSP representation: bandpass, STFT, and log frequency



Bandpass

50–10,000 Hz, constructed as the difference between high-cut and low-cut low-pass FIRs. It reuses the resampler's FIR design and FFT convolution.

STFT

46 ms Hann windows, 12 ms hop (about 74% overlap). Longer windows sharpen frequency but blur time. Hann tapering reduces edge-discontinuity leakage.

Log representation

$20 \log_{10} |X|$ prevents a few loud bins dominating. A 128-bin triangular, $\log_2$ -spaced filterbank makes pitch multiplication become an additive vertical shift.

Resolution caveat

Zero-padding the FFT creates smoother interpolation, not new frequency resolution; true resolution is fixed by the analysis-window duration.

Peak picking

`maximum_filter` finds local maxima in frequency-time neighborhoods. A peak survives only if it passes:

- ▶ global floor: within 40 dB of the clip maximum;
- ▶ adaptive row floor: 15 dB above that row's median;
- ▶ zoning cap: limited peaks per frequency-time region.

The adaptive floor resists steady narrowband interference; zoning preserves quieter call segments.

Pair hashing

Each anchor connects to up to `fan_out` later peaks inside a target-time zone. Pairs are distinctive; individual peaks collide too often.

$$h = \text{pack}(f_1, f_2, \Delta t).$$

Pitch-invariant mode instead uses

$$h = \text{pack}(f_2 - f_1, \lfloor \Delta t / 2 \rfloor),$$

trading extra collisions for pitch and mild stretch tolerance.

DSP matching: offset-histogram voting



For every shared hash and reference occurrence,

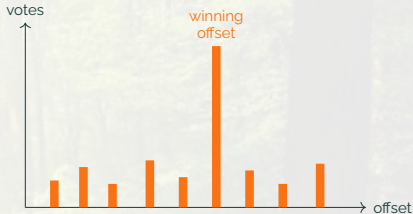
$$\Delta = t_{\text{query}} - t_{\text{reference}}.$$

The matcher counts votes by (reference, Δ). A true occurrence aligns many hashes at one offset; accidental collisions scatter across offsets.

Score

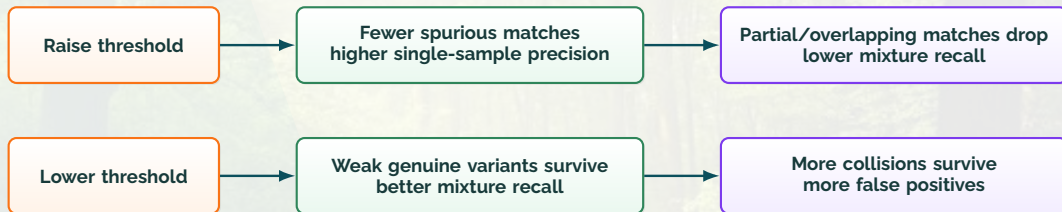
$$\text{score} = \frac{\text{votes at winning offset}}{\text{reference's total hash count}}.$$

It is an uncalibrated overlap fraction, not a probability.



Raw shared-hash count alone is insufficient: temporal agreement is the discriminating signal.

The central DSP trade-off: min_raw_count



Production decision

MIN_RAW_MATCH_COUNT = 5: deliberately low so pitch/tempo variants and partial events survive long enough to vote. The choice favors multi-event robustness over maximum single-clip precision.

Reference variants

Each reference expands over ± 2 semitones and rates 1.0/1.15 (six pitch-tempo variants). Variants score independently; only afterward does `collapse_variant_scores` retain the best variant per species. Premature merging would dilute a match that resembles one variant.

Window aggregation

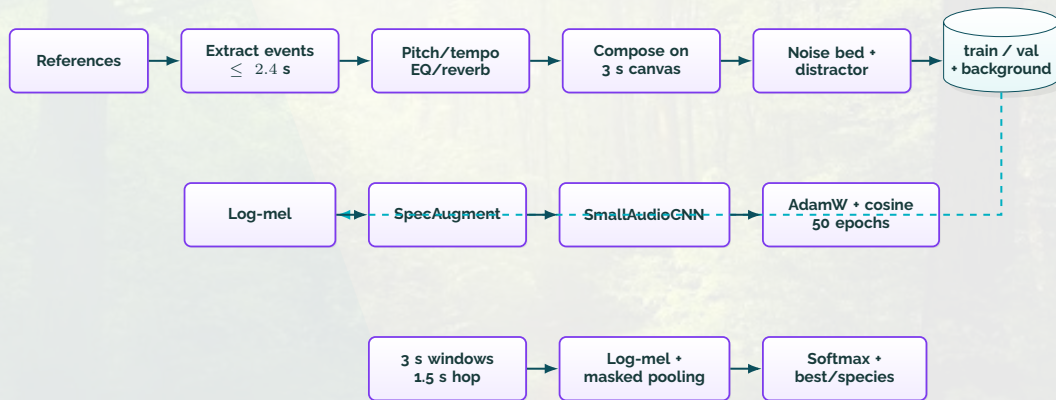
Keep the maximum per species across query windows and stored references. Averaging would penalize a real call that resembles only one reference or occurs in only one window.

`dsp_diagnostics.py`

- ▶ flags suspiciously low reference hash counts;
- ▶ reports hashes per second to separate short clips from corrupt inputs;
- ▶ explains occasional self-match scores above 1: repeated query hashes create an $n \times m$ cross-product against self-similar references.

Output: ranked {species, confidence} at 22,050 Hz mono; confidence is uncalibrated fingerprint overlap.

Pathway 2: trained phase-1 CNN



Why simple crop-or-pad failed



Shortcut learned by the model

References span 0.6–8 s. Zero-padding a 0.6 s bark to 3 s makes about 80% of the input digital silence, while a long crow clip fills the canvas. The CNN can identify “amount of silence” instead of the call.

Measured evidence

The same test sounds scored 88% with zero-padding but only 48% after replacing silence with a realistic noise floor; cat, cow, and dog fell to 0%. The drop exposed leakage, not a deployment-safe model.

Trim + split events
merge gaps < 0.3 s

Augment actual event
not padding

Place randomly; edge cuts;
repeat short calls 1–3×

Fill canvas with realistic
background at 3–35 dB SNR

Synthetic dataset and shared noise generator



Balanced event composition

- ▶ Each class receives exactly `-variants-per-class` clips (typically 200), independent of source duration.
- ▶ Event transforms: pitch ± 3 st, tempo $0.8\text{--}1.25\times$, band-limiting EQ, and synthetic reverb.
- ▶ Random location, occasional edge truncation, call repetition, and optional quiet call from another species.
- ▶ `_background` uses the same generator, so "nothing" belongs to the same acoustic world.

noise_bed.py

White, pink ($1/\sqrt{f}$), brown, mains hum with harmonics, amplitude-modulated low-pass wind, and near-silent quiet. Beds are normalized to unit RMS, then scaled to the requested SNR.

No tiling

Repeating short audio produces autocorrelation near 0.9 versus roughly 0 for real audio—an artificial periodic shortcut. Tiling was rejected in both noise generation and dataset composition.

Input representation

3 s at 22,050 Hz $\rightarrow$ 64-bin log-mel spectrogram using $n_{\text{fft}} = 1024$, hop 256, and 50–10,000 Hz; fixed frame count and scaling approximately to $[-1, 1]$.

SmallAudioCNN

Conv(1 $\rightarrow$ 16)-BN-ReLU-Pool
 $\rightarrow$ Conv(16 $\rightarrow$ 32)-BN-ReLU-Pool
 $\rightarrow$ Conv(32 $\rightarrow$ 64)-BN-ReLU
 $\rightarrow$ masked global average pool
 $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64 $\rightarrow$ classes)

Masked pooling

Short uploads require noise-bed padding. Convolutions and BatchNorm still process it, but `valid_frames` gives padded columns exactly zero weight in global average pooling. This is a structural guarantee, not a statistical hope.

Training samples are unmasked because their entire 3 s canvas is intentionally generated content plus realistic background. The network remains small because roughly one real recording per class, even heavily augmented, cannot support a large model without rapid overfitting.

Training recipe

Epochs	50
Optimizer	AdamW; weight decay 10^{-2}
Schedule	Cosine annealing
Loss	Cross-entropy; smoothing 0.1
Batch / LR	16 / 10^{-3}
Regularization	Dropout + SpecAugment
Validation	15%; best accuracy, then loss

Honest validation

With at least three recordings per class, whole recordings are held out. With fewer, train and validation use augmentations of the same recordings; the resulting optimistic estimate is detected and logged per class. More independent real recordings are the largest accuracy lever.

Inference

Up to 120 s; 3 s windows with 1.5 s hop; best probability per species. A background winner or top score below 0.20 returns no confident detection, preventing closed-set softmax from always inventing a class.

Pathway 3: PANN zero-shot tagging



What it contributes

CNN14 was pretrained on roughly two million AudioSet clips spanning 527 general sound classes. It is used without fine-tuning, providing broad prior knowledge independent of Auralis references and synthetic training.

Documented exception

The pretrained mel front end is inseparable from the learned weights; replacing it would defeat the purpose of transfer. PANN is therefore an explicitly exploratory black-box path, while its required 32 kHz resampling remains hand-built.

Zero-padding of short trailing windows may soften scores because CNN14 exposes no masking hook; this is accepted for the exploratory pathway. Sigmoid output is genuinely multi-label.

Why mapping is necessary

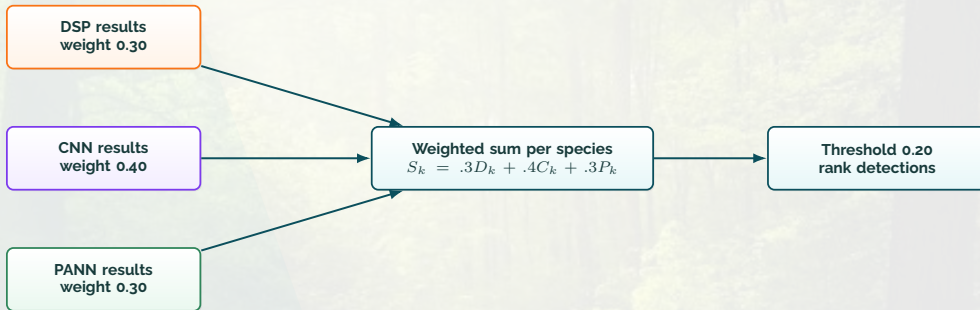
Raw AudioSet labels are both redundant (meow, cat, caterwaul) and out of scope (e.g., rain or music). They cannot be treated as species predictions or merged directly with the other paths.

Mapping rules

- ▶ Synonyms map to one category: bark/dog/bow-wow → dog.
- ▶ Combine synonyms by maximum, never sum, to avoid correlated inflation.
- ▶ Use `farm_animals` only if no specific farm species clears threshold.
- ▶ Suppress generic `animal/bird/noise` tags once a specific category fires; retain as fallback otherwise.
- ▶ Preserve unmapped tags for diagnostics, but exclude them from the ensemble.

Output: relabeled, ranked multi-label detections; minimum PANN score 0.15.

Ensemble: independent failures, shared decision



Robust execution

`process_ensemble_sample()` gives CNN and PANN separate temporary-file copies while DSP reads independently, preventing one consumer from leaving another at end-of-file. Every path is called defensively.

Common contract

All paths return `{species, confidence}` and apply the common 0.20 backend threshold. PANN is restricted to mapped Auralis categories. Missing checkpoints or other failures appear in `path_errors`; remaining paths still return.

The backend returns per-path predictions; the client applies the displayed weights.

Reproducible generated test set



Eight stress variants per species

Original; padded silence; white noise at 10 dB SNR; pitch $+3/ - 3$ st; speed $0.8\times/1.25\times$; and a fixed -12 dB, 1 kHz tone to exercise the adaptive row floor.

Mixtures

Five non-overlapping concatenations and five randomized overlapping mixtures test multi-event aggregation. The first mixture of each kind always includes gunshot.

Ground truth, not filename guesses

`generate_test_samples.py` uses a fixed seed. `generated_manifest.json` records source, transform, and exact event timestamps for every mixture.

Per-path harnesses

`test_dsp.py`, `test_cnn.py`, and `test_pann.py` cache raw JSON and chart single-event top-1 plus mixture confidence/species recovery, separating overlap regimes.

Evaluation: defensible conclusions



Path	Observed qualitative pattern	Expected limitation
DSP	Strong on clean/original clips (often 0.8–1.0); moderate pitch, tempo, and noise tolerance from invariants and variants	Heavy transforms can produce no detection; mixtures often recover only a subset, consistent with the raw-vote threshold trade-off
CNN	Generally confident top-1 across most generated variants (often ≥ 0.8)	Harder transforms may fall below threshold; validation is optimistic when independent recordings are unavailable
PANN	Recognizes several species with no project-specific training	Misses or weakens out-of-distribution calls and project mixtures; labels require careful mapping

Reporting policy

Exact accuracy changes when `samples/test/` is regenerated. Presenting the fixed-seed charts and qualitative cross-path pattern is more defensible than one unstable headline number.

Reference hygiene and operational safeguards

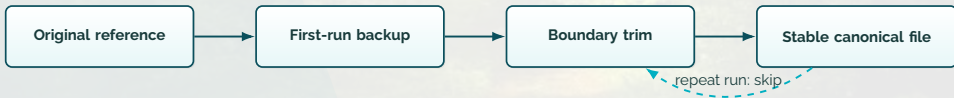


`trim_references.py`

Trims only leading/trailing silence in place and backs up the untouched original on the first run. Later runs are no-ops, preventing compounding lossy re-encoding.

Container handling

Because the writer cannot encode M4A, trimmed M4A references become WAV and the original moves to backup. The dataset builder therefore cannot double-count both versions.



What is technically distinctive



- ▶ Polyphase identity fixes a real OOM without replacing the hand-built resampler.
- ▶ Log-frequency relative hashing converts pitch scaling into translation invariance.
- ▶ Offset voting distinguishes coherent occurrences from coincidental collisions.
- ▶ The `min_raw_count` setting exposes an explicit precision-mixture-recall decision.
- ▶ Masked pooling gives padding provably zero classifier weight.
- ▶ The 88% to 48% experiment uncovered silence leakage and motivated realistic composition.
- ▶ Tiling was independently rejected in two subsystems because of artificial periodicity.
- ▶ Generated tests carry exact, reproducible per-event ground truth.
- ▶ Validation labels itself optimistic when recording-level holdout is impossible.
- ▶ PANN's black-box boundary and required relabeling are explicit, while resampling stays hand-built.

One-page comparison



Aspect	DSP	CNN	PANN
Core idea	Peak-pair hashing + offset votes	Learned log-mel classifier	Pretrained CNN14; relabeled
Training Representation	References only Bandpassed log-frequency STFT	Synthetic data; 50 epochs 64-bin log-mel	None Internal pretrained mel front end
Invariance	Relative pitch, mild tempo, variant grid	Augmentation + SpecAugment	AudioSet coverage
Multi-event	Best/species votes/windows over	3 s sliding windows	6 s multi-label windows
Main dial	min_raw_count	confidence/background threshold	confidence + label mapping
Output Weight	Uncalibrated overlap 0.30	Softmax probability 0.40	Relabeled sigmoid 0.30

Bottom line

The three paths fail differently: exact-pattern evidence, project-specific learned similarity, and broad pretrained knowledge. Shared preprocessing makes them comparable; defensive execution and a common vocabulary make them mergeable.

Appendix: input/output and tunables



	DSP	CNN	PANN
Input limit	One uploaded file / whole clip	Up to 120 s	Up to 120 s
Preprocess	22,050 Hz mono; 50–10k; 46/12 ms; 128 log bins	22,050 Hz; 3/1.5 s; 64 mels	32,000 Hz; 6/3 s
Reference/data	samples/references/*; six variants	200/class + _background	AudioSet pretraining
Threshold	common 0.20 after DSP scoring	MIN_CONFIDENCE=0.20	internal 0.15; common 0.20 for ensemble
Key controls	fan-out, target Δt , pitch invariance, time bucket, minimum votes	augmentation, SpecAugment, masked pooling, background class	mapping, fallback suppression, confidence

Samples for Benchmark



Single-event: 88

cat_noised.wav
cat_original.wav
cat_padded_silence.wav
cat_pitch_decreased.wav
cat_pitch_increased.wav
cat_slowed.wav
cat_speeded.wav
cat_toned.wav
cow_noised.wav
cow_original.wav
cow_padded_silence.wav
cow_pitch_decreased.wav
cow_pitch_increased.wav
cow_slowed.wav
cow_speeded.wav
cow_toned.wav
crow_noised.wav
crow_original.wav
crow_padded_silence.wav
crow_pitch_decreased.wav
crow_pitch_increased.wav
crow_slowed.wav
crow_speeded.wav
crow_toned.wav
dog_noised.wav

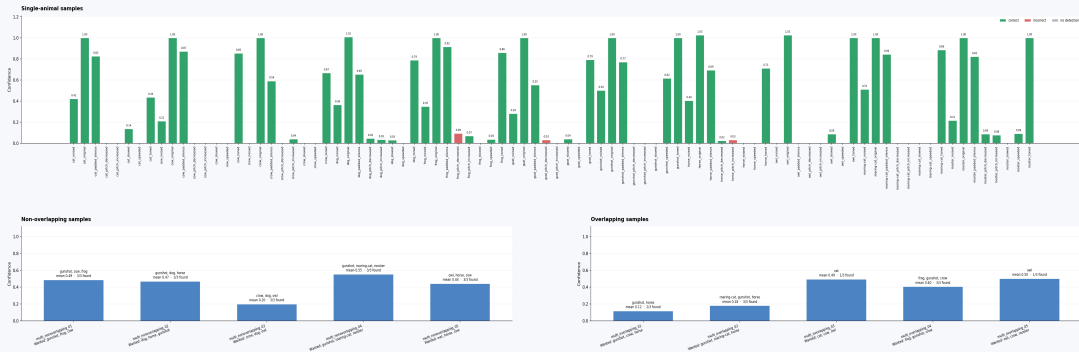
dog_original.wav
dog_padded_silence.wav
dog_pitch_decreased.wav
dog_pitch_increased.wav
dog_slowed.wav
dog_speeded.wav
dog_toned.wav
frog_noised.wav
frog_original.wav
frog_padded_silence.wav
frog_pitch_decreased.wav
frog_pitch_increased.wav
frog_slowed.wav
frog_speeded.wav
frog_toned.wav
goat_noised.wav
goat_original.wav
goat_padded_silence.wav
goat_pitch_decreased.wav
goat_pitch_increased.wav
goat_slowed.wav
goat_speeded.wav
goat_toned.wav
gunshot_noised.wav
gunshot_original.wav

Non-overlapping: 5

gunshot_padded_silence.wav
gunshot_pitch_decreased.wav
gunshot_pitch_increased.wav
gunshot_slowed.wav
gunshot_speeded.wav
gunshot_toned.wav
horse_noised.wav
horse_original.wav
horse_padded_silence.wav
horse_pitch_decreased.wav
horse_pitch_increased.wav
horse_slowed.wav
horse_speeded.wav
horse_toned.wav
owl_noised.wav
owl_original.wav
owl_padded_silence.wav
owl_pitch_decreased.wav
owl_pitch_increased.wav
owl_slowed.wav
owl_speeded.wav
owl_toned.wav
roaring-cat_noised.wav
roaring-cat_original.wav
roaring-cat_padded_silence.wav

Overlapping: 5

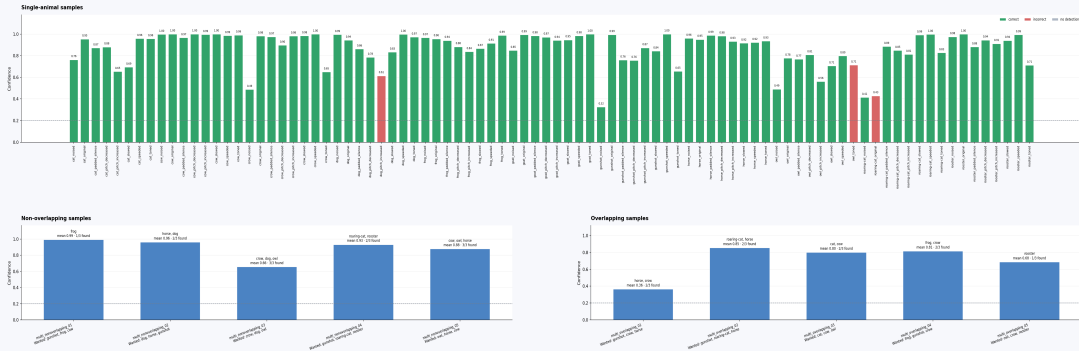
roaring-cat_pitch_decreased.wav
roaring-cat_pitch_increased.wav
roaring-cat_slowed.wav
roaring-cat_speeded.wav
roaring-cat_toned.wav
rooster_noised.wav
rooster_original.wav
rooster_padded_silence.wav
rooster_pitch_decreased.wav
rooster_pitch_increased.wav
rooster_slowed.wav
rooster_speeded.wav
rooster_toned.wav
multi_nonoverlapping_01.wav
multi_nonoverlapping_02.wav
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multi_overlapping_04.wav
multi_overlapping_05.wav

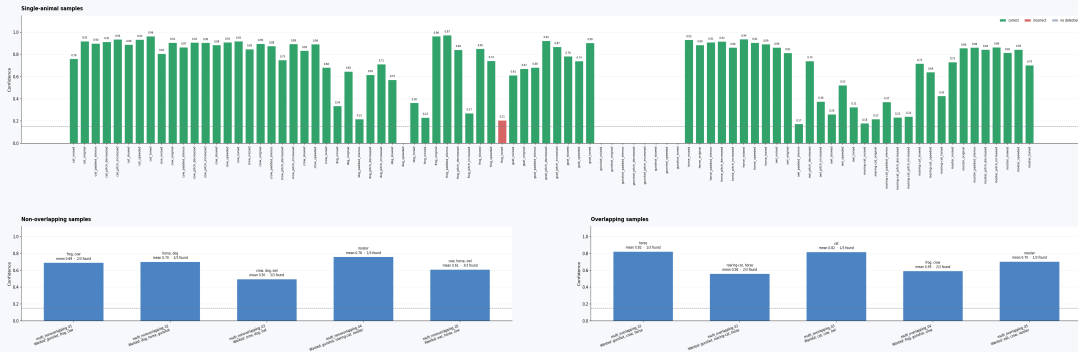


Benchmark: Trained CNN



Auralis CNN results





Auralis

Three complementary signal analysis techniques. One decision.

DSP FINGERPRINTS · TRAINED CNN · PRETRAINED PANN
